Beyond Additive Symptom Scores: A Bivariate Probabilistic Instrument for Gastroesophageal Reflux Disease

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August 2026

Abstract

Symptom questionnaires for gastroesophageal reflux disease (GERD) commonly convert frequency and intensity into ordinal points and add them. This is convenient but imposes strong, usually untested assumptions: adjacent categories are equally spaced; one unit of frequency is exchangeable with one unit of intensity; symptom effects are linear; interactions are absent; and one threshold is transportable across populations. We propose a theoretical replacement for arithmetic scoring, the Bivariate Reflux Probability Instrument (BRPI). BRPI retains symptom frequency and intensity as separate dimensions and estimates the probability of objectively supported, actionable GERD. Its preferred architecture has two layers: a bivariate graded-response measurement model that estimates latent frequency and intensity while accounting for reporting uncertainty, followed by a monotone Bayesian generalized additive model that learns nonlinear symptom-specific surfaces and interactions. Heartburn and regurgitation are core indicators; nocturnal symptoms and symptom-related sleep disturbance may add information, while epigastric pain and nausea can act as differential-diagnosis features. Alarm symptoms bypass the score and trigger clinical assessment. The output is not a total score but a calibrated probability with uncertainty and three action zones: low probability, indeterminate, and high probability. To demonstrate clinical use, we provide a one-page sample questionnaire, a completed patient example, and an automatically generated probability report with an uncertainty interval, action zone, explanatory drivers, and patient-specific curve. Coefficients and thresholds must be learned prospectively against a prespecified reference standard based on contemporary endoscopy and reflux-monitoring criteria, with internal, temporal, geographic, and impact validation. Fuzzy membership functions can improve the response interface, but fuzzy rules should not substitute for outcome-calibrated statistical learning. BRPI is intended to replace crude addition within symptom-based triage, not to replace endoscopy or ambulatory reflux monitoring when objective confirmation is clinically required.

Keywords: gastroesophageal reflux disease; GERD; questionnaire; fuzzy logic; item response theory; Bayesian model; generalized additive model; diagnostic probability; calibration; clinical prediction model

1 Introduction

GERD is defined by troublesome symptoms or complications caused by reflux of gastric contents, but symptoms, mucosal injury, reflux burden, and symptom perception do not coincide perfectly [1–5]. Heartburn and regurgitation increase the prior probability of GERD, yet similar complaints occur in reflux hypersensitivity, functional heartburn, dyspepsia, achalasia, rumination, cardiac disease, and other conditions [6–9]. Contemporary guidance therefore distinguishes symptom-based triage from objective confirmation by endoscopy and ambulatory reflux monitoring [2, 3, 6, 10, 11].

Questionnaires such as the Reflux Disease Questionnaire (RDQ), GerdQ, FSSG, and ReQuest have improved standardization and are useful for symptom measurement [12–18]. Nevertheless, their usual arithmetic form can obscure structure. If frequency is coded 0–3 and intensity 0–3, addition treats a severe symptom once monthly as equivalent to a mild symptom on most days whenever the point totals match. It also assumes that the transition from "never" to "one day" has the same meaning as the transition from "four days" to "daily," and that intensity and frequency contribute independently. Meta-analytic evidence indicates that commonly used symptom instruments provide only moderate diagnostic accuracy, which is consistent with both imperfect measurement and biological overlap [14, 15, 19–22].

The mathematical objection is therefore valid, but the solution is not simply to replace addition with a more complicated formula. The model must be trained against a clinically defensible outcome, remain calibrated in the intended setting, quantify uncertainty, and improve decisions rather than only the area under a receiver operating characteristic curve. This article proposes BRPI as a testable methodology. It combines psychometrics, nonlinear probability modeling, and evidence-based diagnostic evaluation. All numerical coefficients and curves below are illustrative until estimated in clinical data.

2 Why a Simple Total Score Is an Inadequate Measurement Model

Let F_j and I_j denote reported frequency and intensity for symptom j. A conventional questionnaire calculates

$$S = \sum_{j=1}^{J} (a_j F_j + b_j I_j), \tag{1}$$

then declares GERD when $S \geq c$. Unless supported by data, equation (1) assumes interval-scaled categories, constant weights, additivity, no interaction, no setting effect, and a sharp biologic boundary at c. These assumptions are rarely evaluated.

Frequency and intensity are related but non-substitutable. Frequency approximates repeated exposure or repeated perception over time; intensity reflects the magnitude of individual episodes, central amplification, mucosal sensitivity, language, and recall. A product $F_j I_j$ is also insufficient: it forces zero burden when either dimension is zero, produces unstable extremes, and still assumes a fixed functional form. A clinically plausible model should allow severe rare regurgitation, mild daily heartburn, and frequent intense symptoms to occupy different locations on a risk surface.

The total-score threshold also discards uncertainty. A probability of 0.49 and 0.51 should not automatically trigger radically different care when measurement error is substantial. Finally, predictive values depend on disease prevalence and referral spectrum, so a cutoff derived in gastroenterology clinics cannot be assumed to work in community screening [23–27].

Table 1: Assumptions hidden by additive questionnaire scores and proposed remedies.

Problem	Consequence	BRPI remedy
Ordinal categories treated as equal intervals	Category spacing is imposed rather than estimated	Graded-response thresholds or continuous day counts and visual analogue intensity
Frequency exchanged for intensity	Equal totals can represent clinically different profiles	Preserve two dimensions and learn a response surface
Linear contribution	Risk may plateau, accelerate, or show thresholds	Monotone splines with shrinkage
No symptom interaction	Co-occurring heartburn and regurgitation may change probability	Prespecified pairwise tensor interactions
Hard cutoff without uncertainty	Borderline responses are overinterpreted	Probability interval and indeterminate zone
One cutoff for every setting	Predictive value and calibration drift with case mix	Setting-specific intercept recalibration and external validation
Symptoms equated with objective GERD	Functional disorders and hypersensitivity are misclassified	Reference-standard testing and optional multiclass phenotyping

3 Intended Use and Target Condition

The intended-use statement must be fixed before model development. We propose adult primary-care or outpatient triage among people with upper gastrointestinal symptoms, before a definitive diagnosis. The target is *actionable GERD*: sufficient objective or clinical evidence that reflux-directed management is justified, defined prospectively using contemporary consensus criteria [2, 3, 6, 7]. BRPI should not be used in children, pregnancy, previously proven severe GERD, post-surgical anatomy, or major motility disease without separate validation.

Alarm features are not score items. Dysphagia, odynophagia, gastrointestinal bleeding, anemia, unintentional weight loss, persistent vomiting, a mass, or concerning chest pain activate a safety pathway regardless of calculated probability. Similarly, chest symptoms requiring urgent cardiac evaluation must not be delayed by a reflux model.

The reference standard should be applied without knowledge of BRPI output and may include endoscopy, prolonged wireless pH monitoring, catheter pH-impedance monitoring, and manometry when needed to exclude major motility disorders [3,6,10,11,28–34]. Conclusive positive and conclusive negative findings should follow a locked protocol. Barrett's esophagus and other complications require their established clinical pathways [35]. Borderline physiology should not be forced into a binary label; it can be adjudicated using blinded expert review or modeled as a latent/intermediate class.

4 Proposed Instrument

4.1 Core symptom observations

For each core symptom, the patient reports frequency and intensity separately over a seven-day window. Frequency should preferably be the number of symptomatic days (0–7) and, when feasible, the number of episodes. Intensity can use an anchored 0–10 scale: 0, none; 3, noticeable without activity limitation; 5, interferes with usual activity; 7, marked limitation; and 10, worst imaginable. Anchors reduce linguistic drift better than unlabelled adjectives.

Table 2: Proposed BRPI data fields. Items are candidates to be reduced after development.

Domain	Measurement	Role
Heartburn	Days/week; episodes/week; 0–10 intensity	Core positive indicator
Acid or liquid regurgitation	Days/week; episodes/week; 0–10 intensity	Core positive indicator; may behave differently from heartburn
Nocturnal reflux symptoms	Nights/week and sleep interruption	Additional burden and timing
Postprandial relation	Proportion of episodes after meals	Contextual modifier
Positional relation	Supine/bending association	Contextual modifier
Epigastric pain and nausea	Separate frequency and intensity	Differential-diagnosis information, not simply reverse-scored
Antacid response	Time-stamped response, if observed	Exploratory feature; avoid circular diagnosis and recall bias
Medication and prior probability	PPI use, age, BMI, pregnancy, prior GERD evidence	Calibration and applicability variables
Alarm features	Present/absent with details	Safety override; excluded from numerical trade-offs
Response certainty	Slider or "unsure" option	Measurement uncertainty/fuzzy interface

Electronic administration can collect a short symptom diary rather than one retrospective answer. Repeated observations distinguish a patient with one memorable severe episode from one with a stable high burden and reduce recall error. Adaptive questioning can stop when posterior uncertainty is already sufficiently narrow, but only after item-response calibration.

4.2 Sample clinical questionnaire and automatic probability report

The questionnaire below shows a short clinic-facing version suitable for a tablet, web form, or paper worksheet. The rightmost column contains a completed fictional example so that the transformation from answers to probability can be followed. In production software, alarm-feature answers are evaluated before the probability model; a positive safety response stops automated triage and displays the appropriate clinical pathway.

BRPI-7 demonstration questionnaire: symptoms during the last 7 days

Patient ID: DEMO-001 **Completion date:** August 17, 2026 **Setting:** outpatient triage

Safety screen—do not calculate a probability when present.
Difficult or painful swallowing, vomiting blood/black stool, unexplained weight loss, persistent vomiting, anemia, a mass, or concerning chest pain? ☐ Yes ☒ No

Question	Response options	Example answer
1. Heartburn frequency	Number of days with burning behind the breastbone: **0 1 2 3 4 5 6 7**	**5 days**
2. Heartburn intensity	0 none; 3 noticeable; 5 interferes with activity; 7 marked limitation; 10 worst imaginable	**6/10**
3. Regurgitation frequency	Number of days with acid or liquid returning toward the throat: **0 1 2 3 4 5 6 7**	**4 days**
4. Regurgitation intensity	Anchored 0–10 intensity scale as above	**5/10**
5. Nocturnal reflux	Number of nights with reflux symptoms or reflux-related awakening: **0–7**	**2 nights**
6. Temporal pattern	After meals: ☐ no ☐ sometimes ☒ most episodes; supine/bending: ☐ no ☐ sometimes ☒ often	**postprandial and positional**
7. Differential symptoms	Epigastric pain: days 0–7 and intensity 0–10; nausea: days 0–7 and intensity 0–10	**pain 1 day, 3/10; nausea 0**
Response certainty	How certain are you that the answers represent a typical week? 0–100%	**80%**
Current context	PPI or acid blocker, previous GERD evidence, BMI, pregnancy, and major comorbidity	**no current PPI**

Submit responses and generate probability report

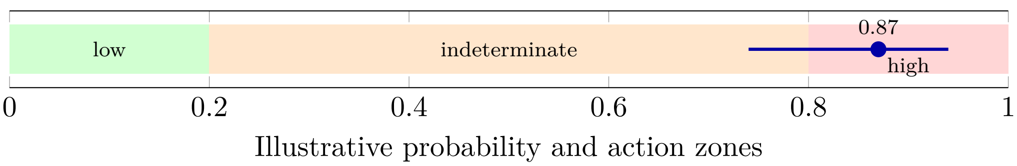
For demonstration only, let H_F, H_I, R_F, R_I denote heartburn and regurgitation frequency and intensity, N nocturnal nights, P postprandial association, S supine/bending association, and E epigastric-pain days. The automatic example uses

$$\begin{aligned}\eta = & -6.60 + 0.30H_F + 0.20H_I + 0.35R_F + 0.24R_I + 0.045H_FH_I + 0.055R_FR_I \\ & + 0.18N + 0.25P + 0.20S - 0.10E, \qquad \widehat{p} = \mathrm{logit}^{-1}(\eta). \end{aligned} \tag{2}$$

For the displayed answers, $(H_F, H_I, R_F, R_I, N, P, S, E) = (5, 6, 4, 5, 2, 1, 1, 1)$, giving $\eta = 1.86$ and $\widehat{p} = 0.87$. The interval below is deliberately illustrative; real uncertainty must be obtained from the fitted posterior distribution and measurement model.

Automatically generated BRPI demonstration report

Estimated probability	**0.87** (illustrative 90% interval 0.74–0.94)
Action zone	**High-probability demonstration zone**; check clinical context and use objective testing whenever required for management.
Main positive drivers	Heartburn on 5 days, regurgitation on 4 days, moderate-to-high intensities, nocturnal symptoms, and postprandial/positional association.
Uncertainty explanation	The interval remains non-negligible because the patient reported 80% response certainty and symptoms vary across days.
Safety statement	This illustrative probability is not proof of GERD and must not delay evaluation of alarm or cardiac symptoms.



The same software can immediately generate a patient-specific curve. In figure 1, the horizontal axis varies heartburn frequency while holding the other completed responses fixed. The red point is the patient's answer. This display explains how changing one response changes probability without implying that heartburn frequency alone causes the prediction.

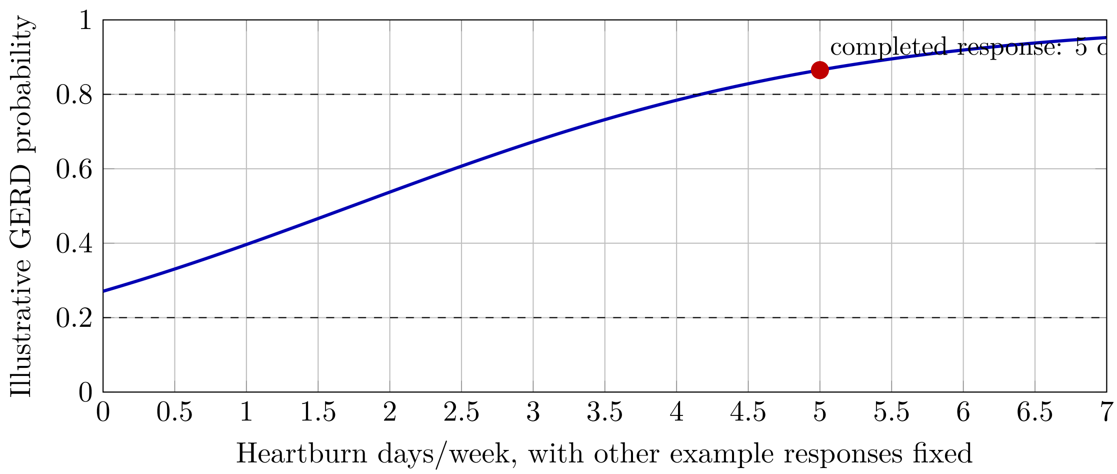


Figure 1: Automatically generated patient-specific demonstration curve. The model varies heartburn frequency and holds all other questionnaire answers fixed. Coefficients and action thresholds are illustrative and are not ready for clinical use.

4.3 Measurement layer: bivariate graded-response model

Ordinal answers should be treated as ordered categories rather than numerical distances. For person n, symptom j, dimension $d \in \{F, I\}$, and response threshold k, a graded-response model is

$$\Pr(Y_{njd} \geq k \mid \theta_{nd}) = \operatorname{logit}^{-1}[a_{jd}(\theta_{nd} - b_{jdk})], \tag{3}$$

where θ_{nF} and θ_{nI} are correlated latent frequency and intensity, a_{jd} is item discrimination, and b_{jdk} is an ordered threshold. This estimates category spacing rather than assuming it. Differential item functioning must be examined by language, sex, age, culture, and care setting [36–44].

If exact day counts and anchored continuous intensity are reliable, the diagnostic layer may use those observations directly. The IRT layer is most valuable when responses remain categorical, when adaptive testing is desired, or when response uncertainty must be modeled. A bifactor or multidimensional model can determine whether a common reflux-symptom trait exists while retaining symptom-specific frequency and intensity.

4.4 Diagnostic layer: monotone Bayesian generalized additive model

For binary actionable GERD D_n, the preferred initial model is

$$\begin{aligned}\operatorname{logit}\Pr(D_n = 1) = \alpha_s &+ f_{HF}(\theta^H_{nF}) + f_{HI}(\theta^H_{nI}) + f_{RF}(\theta^R_{nF}) + f_{RI}(\theta^R_{nI}) \\ &+ g_H(\theta^H_{nF}, \theta^H_{nI}) + g_R(\theta^R_{nF}, \theta^R_{nI}) + \boldsymbol{\beta}^{\mathsf{T}}\mathbf{x}_n,\end{aligned} \tag{4}$$

where H denotes heartburn, R regurgitation, f are monotone smooth functions, g are shrinkage-controlled tensor interactions, $\mathbf{x}$ contains prespecified modifiers, and α_s is a setting-specific intercept. Priors constrain implausible oscillation and excessive interaction. The model outputs a posterior distribution for probability rather than one deterministic score [45–50].

Monotonicity is desirable for core symptoms: holding other variables constant, greater frequency should not lower predicted GERD probability unless strong evidence supports a non-monotone relation. Negative or differential-diagnosis items need not be reverse-scored; their learned functions may be weak, nonlinear, or context dependent. The full model should be compared with a simpler penalized logistic model. Complexity is retained only if it improves calibration and net benefit in validation.

4.5 Optional fuzzy response layer

Fuzzy mathematics is useful for translating vague words into overlapping memberships. For example, frequency $f \in [0, 7]$ can have memberships in "rare," "intermittent," and "frequent" sets. A smooth logistic membership is

$$\mu_{\text{frequent}}(f) = [1 + \exp\{-\kappa(f - \tau)\}]^{-1}. \tag{5}$$

Intensity can be represented similarly, and a patient may belong partly to adjacent categories. This avoids an artificial jump between "2" and "3" days. Mamdani or Takagi–Sugeno systems can combine linguistic rules [51–55].

However, expert-written rules such as "IF frequency is high AND intensity is severe THEN GERD is likely" do not guarantee diagnostic validity. Membership functions and rule consequents should be learned or recalibrated against reference outcomes. For this reason, BRPI uses fuzzy membership primarily as a response and explainability layer; the final probability is outcome-calibrated. A purely fuzzy system is a secondary comparator, not the default.

5 Illustrative Probability Curves

To make the proposal concrete, consider the deliberately simplified function

$$\widehat{p} = \operatorname{logit}^{-1}\left[-4.2 + 0.52F + 0.38I + 0.075FI\right], \tag{6}$$

where F is symptomatic days per week and I is intensity from 0 to 10. These coefficients are not clinical estimates. They only demonstrate that frequency changes the implication of intensity and vice versa.

The curves show why equal sums need not be equivalent. For example, $(F, I) = (6, 2)$ and $(2, 6)$ both sum to 8, but equation (6) assigns different probabilities because repeated low-intensity symptoms and rare high-intensity symptoms carry different patterns. In the final instrument, separate surfaces would be estimated for heartburn and regurgitation and combined with other validated predictors.

6 From Probability to a Clinical Action

BRPI should display: (i) estimated probability; (ii) a 90% or 95% uncertainty interval; (iii) the most influential observations in plain language; and (iv) the recommended next step. A three-zone policy is safer than one threshold:

- **Low probability:** consider alternative diagnoses and conservative management, while respecting clinical judgment.
- **Indeterminate:** repeat diary measurement, assess confounders, or perform objective testing when the result changes management.
- **High probability:** a structured reflux-directed pathway may be reasonable in an uncomplicated presentation; objective evidence remains necessary before invasive treatment or when long-term diagnostic certainty matters.

Thresholds must be selected using consequences. If missing significant disease is much more harmful than unnecessary testing, the lower threshold is reduced. For threshold probability p_t, decision-curve net benefit is

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}}{N} - \frac{\mathrm{FP}}{N}\frac{p_t}{1 - p_t}. \tag{7}$$

The model should be adopted only if it has higher net benefit than “test all,” “test none,” and current questionnaires across a clinically relevant threshold range [56–58].

7 Development Study

7.1 Design and population

A prospective, multicenter, cross-sectional diagnostic study should enroll consecutive symptomatic adults before the reference standard. Primary care, general gastroenterology, and physiology referral settings should be sampled explicitly. Spectrum exclusions should be minimized and documented. Participants complete the electronic diary without seeing their probability; reference-test assessors remain blinded to questionnaire output.

The reference protocol should define proven GERD, evidence against GERD, and inconclusive evidence using current endoscopic and reflux-monitoring criteria [2, 3, 6, 10, 29]. Verification bias

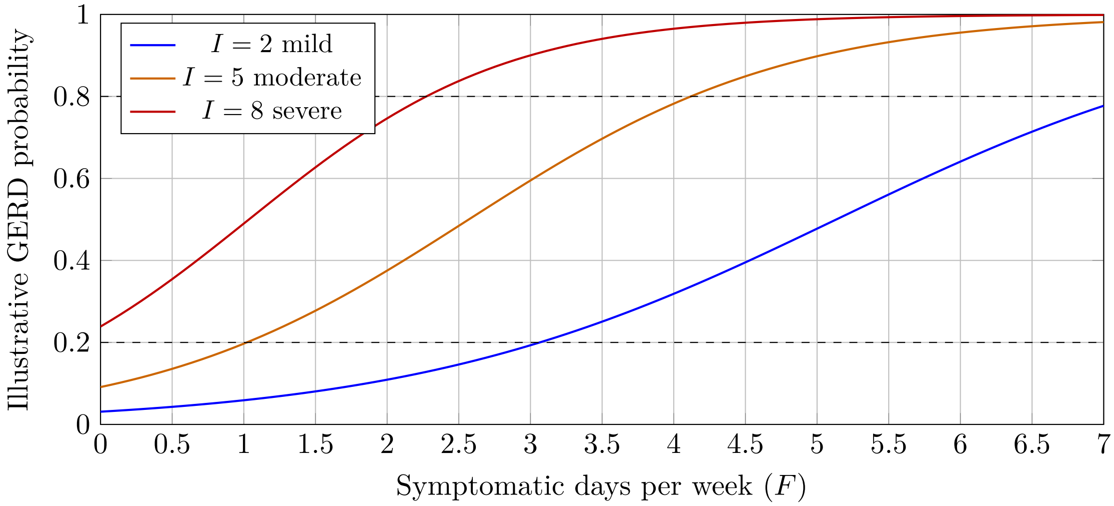


Figure 2: Illustrative probability curves preserving frequency and intensity as separate dimensions. Dashed lines show hypothetical action thresholds, not validated clinical cutoffs.

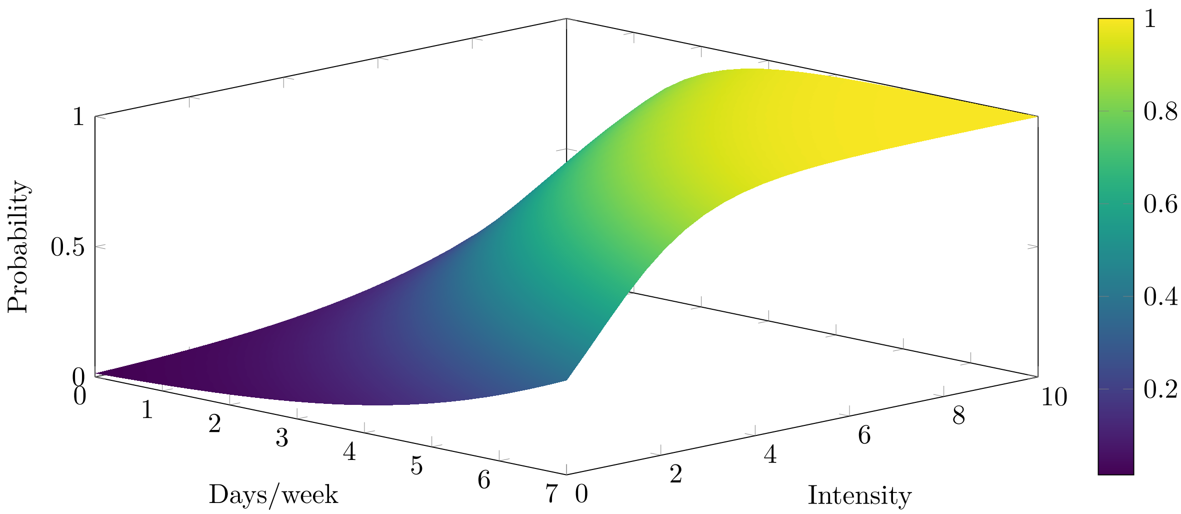


Figure 3: Illustrative bivariate probability surface. The curvature and interaction must be learned from data; the figure is a design example rather than a diagnostic calculator.

occurs if only high-scoring patients receive monitoring, so either all participants need an appropriate reference assessment or sampling weights and sensitivity analyses must address partial verification. Treatment with proton pump inhibitors, test performed on or off therapy, and prior proof of GERD must be prespecified.

7.2 Sample size and model specification

Sample size should be based on anticipated event fraction, number of effective parameters, expected model fit, desired shrinkage, and precision of calibration rather than a simplistic events-per-variable rule [45–47]. Flexible surfaces consume more degrees of freedom than linear terms. A practical development program would likely require several thousand participants distributed across centers, with enough conclusive positive and negative cases for precise calibration and subgroup assessment.

All candidate predictors, transformations, interactions, priors, and primary performance measures should be specified before outcome access. Continuous variables should remain continuous. Missing responses should be minimized by interface design and handled by principled multiple imputation, with sensitivity analyses for informative missingness [59,60].

7.3 Model fitting and internal validation

Penalized maximum likelihood or Bayesian shrinkage should control overfitting. Bootstrap validation estimates optimism in the entire modeling procedure, including feature selection, smoothing, and calibration. If machine-learning comparators are evaluated, hyperparameter tuning must be nested within resampling; otherwise performance is optimistic [61–65]. Random forests and boosting can be useful comparators, but they do not automatically outperform regression and may be less well calibrated [66]. If non-probabilistic scores are retained, Platt scaling or isotonic calibration must be evaluated without leaking validation outcomes into development [67,68].

The following models should be compared prospectively:

- current arithmetic questionnaire;
- linear penalized logistic regression using separate frequency and intensity;
- monotone GAM with limited interactions;
- bivariate IRT plus monotone GAM;
- calibrated fuzzy inference system;
- one carefully tuned machine-learning benchmark.

8 Performance Evaluation

Discrimination alone is insufficient. The primary metrics should include calibration-in-the-large, calibration slope, a smooth calibration curve, Brier score, log loss, sensitivity and specificity at prespecified action thresholds, and decision-curve net benefit [26,58,69–72]. The area under the curve measures ranking but cannot show whether a reported 70% risk is actually 70%. Reclassification indices can be secondary and must not replace clinically interpretable changes [73].

External validation should report performance before and after recalibration. Updating may require only a new intercept in a population with different prevalence, or it may require coefficient and shape revision when symptom effects differ. Performance should be evaluated across sex, age,

Table 3: Minimum validation framework for BRPI.

Validation stage	Purpose	Required evidence
Internal	Estimate overfitting and tune complexity	Bootstrap optimism correction; full-pipeline resampling
Temporal	Test performance after time-related changes	New consecutive cohort; calibration and decision curves
Geographic	Test transportability between centers/countries	Diverse prevalence, languages, ethnicity, and referral spectra
Measurement invariance	Ensure items behave comparably	Differential item functioning and response-process interviews
Clinical utility	Determine whether decisions improve	Net benefit, test utilization, missed important disease, patient burden
Impact trial	Compare model-guided care with usual care	Cluster or individual randomization; safety, outcomes, costs, acceptability
Post-deployment	Detect drift and unequal performance	Calibration monitoring, subgroup auditing, version control, periodic updating

language, health literacy, BMI, ethnicity, PPI exposure, erosive and non-erosive disease, and care setting. Subgroup sample sizes and uncertainty must be reported; apparent differences from small samples should not be overinterpreted [27, 74, 75].

9 Multiclass Extension

A binary GERD label may be too coarse. A clinically richer version estimates probabilities for: (1) objective pathological reflux; (2) reflux hypersensitivity; (3) functional heartburn or another disorder; and (4) inconclusive evidence. Multinomial or hierarchical Bayesian modeling can combine symptoms with pH-impedance metrics while preserving uncertainty. This extension reflects the fact that symptom intensity can be high despite physiological acid exposure within reference limits [3, 8, 9, 76–79].

The multiclass model is preferable for specialist use, whereas a three-zone binary output may be easier in primary care. These are different intended uses and require separate validation. The label must never be inferred from symptom response to proton-pump inhibitors alone because treatment response has limited specificity and can introduce incorporation bias [80–82].

10 Human Factors and Cross-Language Development

The instrument should be developed as a patient-reported outcome measure, not merely translated word for word. Cognitive interviews should test comprehension of heartburn, acid regurgitation, intensity anchors, day counts, and uncertainty. Forward translation, reconciliation, back translation, expert review, and patient testing are followed by measurement-invariance analysis [41–44, 83, 84].

An interface can display a two-dimensional symptom grid: horizontal movement records days per week and vertical movement records intensity. A translucent ellipse around the selected point can represent response uncertainty. This is a useful fuzzy representation because patients need not pretend to know an exact value. The statistical model integrates over the reported uncertainty instead of converting it into a false precise integer.

Explanations should be concise: "Your probability is driven mainly by frequent regurgitation and nocturnal symptoms; the uncertainty is wide because symptoms varied over the week." Feature-attribution methods may support explanation, but they do not establish causality and should be checked for stability [85]. Accessibility, low-literacy use, screen-reader compatibility, and a paper fallback are required.

11 Bias, Safety, and Regulatory Considerations

Major threats include selection bias, differential verification, incorporation of the index test into diagnosis, unblinded adjudication, missing reference tests, overfitting, data leakage, and post hoc cutoff selection. Diagnostic studies should follow STARD, prediction-model reports should follow TRIPOD, and bias should be assessed with QUADAS-2 and PROBAST or their current applicable extensions [86–92]. If an adaptive or artificial-intelligence-enabled version is tested prospectively, protocol and trial reports should follow SPIRIT-AI and CONSORT-AI [93–96].

Clinical software must preserve model version, input units, applicable population, calibration date, uncertainty, and audit trails. Updating should be governed rather than silently continuous. The output must state that a probability is not proof of disease. High-risk safety symptoms must always override the prediction.

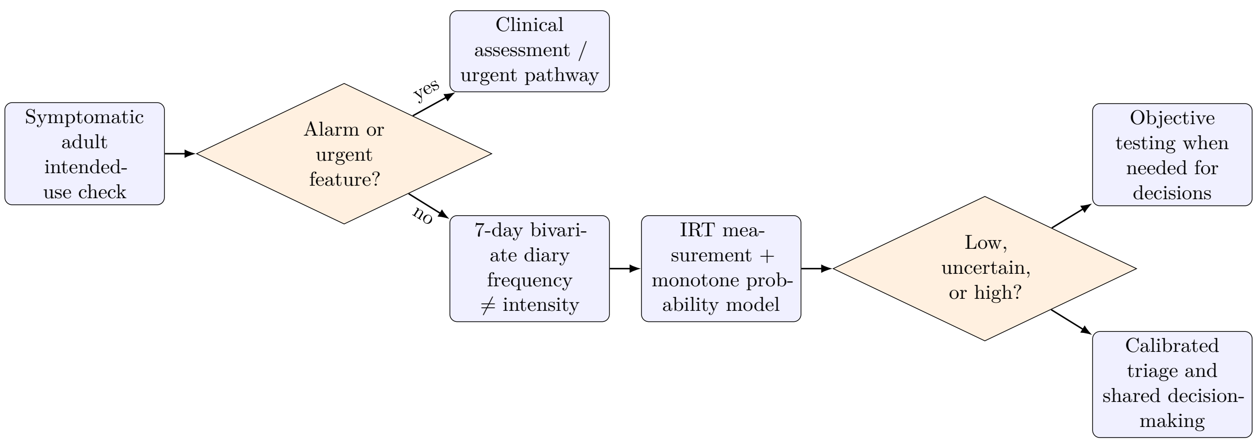


Figure 4: Proposed clinical workflow. BRPI replaces arithmetic symptom addition within triage; it does not override urgent assessment or objective confirmation.

12 Proposed Research Program

- **Qualitative development:** interviews and international Delphi work define symptoms, anchors, uncertainty responses, and intended use.
- **Pilot measurement study:** repeated diary data estimate reliability, response distributions, dimensionality, and differential item functioning.
- **Prospective diagnostic development:** consecutive recruitment with blinded reference assessment and a locked analysis plan.

- **External validation:** temporal and geographic cohorts, including a Chinese-language cohort developed through cultural adaptation rather than literal translation.
- **Model reduction:** derive the shortest instrument that preserves calibration and net benefit; publish coefficients and a transparent calculator.
- **Impact trial:** compare BRPI-guided care with GerdQ/RDQ or usual practice, measuring objective-test use, appropriate treatment, missed alternative disease, symptoms, costs, and patient experience.
- **Surveillance:** monitor drift, adverse consequences, subgroup calibration, and changes in testing standards.

Data and code should be shared when governance permits. A frozen external test set should be protected from iterative development. Negative findings—for example, no improvement over a simple logistic model—are scientifically important. A complex model is justified only by reproducible clinical utility.

13 Discussion

The proposed method answers the central mathematical concern: frequency and intensity should not be added merely because both are represented by small integers. They are distinct measurements with potentially nonlinear and interacting associations with disease. A bivariate probability surface is more faithful than a sum, while a graded-response layer respects the ordinal and uncertain nature of patient reports.

Fuzzy logic is useful but not sufficient. It elegantly represents vague boundaries and may make the interface more natural; however, an expert rule base can be confidently wrong. Outcome-calibrated regression, GAMs, IRT, and Bayesian uncertainty provide a stronger primary framework. Modern machine learning may be explored, but flexibility must not be confused with evidence, and calibration is more important than algorithmic novelty [45, 46, 48, 49, 63, 66, 69].

The largest limitation is the target itself. Symptoms cannot perfectly recover reflux physiology because GERD is heterogeneous and perception is influenced by sensitivity and gut–brain interaction. Therefore BRPI can plausibly replace crude arithmetic questionnaires as a triage and probability instrument, but it cannot responsibly replace endoscopy or reflux monitoring in every patient. Its success should be judged by better decisions and patient outcomes, not by whether it produces an attractive curve.

14 Conclusion

A mathematically defensible GERD symptom tool should preserve frequency and intensity as separate dimensions, learn nonlinear symptom-specific relationships against a prespecified reference standard, report calibrated probability and uncertainty, and link predictions to explicit clinical actions. BRPI combines a bivariate graded-response model, monotone Bayesian generalized additive modeling, optional fuzzy response representation, and three-zone decision analysis. The proposed equations and figures are theoretical, not ready-to-use diagnostic thresholds. Prospective multicenter development, external validation, cross-language measurement testing, and an impact trial are required before clinical replacement of existing questionnaires.

Declarations

Ethics approval: Not applicable to this theoretical methodological article. **Funding:** No funding was declared. **Conflicts of interest:** The authors should disclose all relevant interests before submission. **Data availability:** No patient-level data were generated. The illustrative coefficients must not be used for clinical diagnosis.

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